

Def. A field extension E/F is called a simple radical extension 'if:

$E = F(\alpha)$ for some α satisfying $\alpha^n \in F$ for some $n \geq 2$.

Moreover, a field extension R/F is called a Radical extension 'if:
there exists a tower $F = F_0 \subset F_1 \subset \dots \subset F_n = R$ such that

F_i/F_{i-1} is a simple radical for each $1 \leq i \leq n$.